MIL-U-36000B
Int. Amendment - 1(DLA-DM)
13 April 1977

MILITARY SPECIFICATION

UNDECYLENIC ACID AND ZINC UNDECYLENATE POWDER

This interim amendment is issued for use by Defense Personnel Support Center with Military Specification MIL-U-36000B dated 2 April 1976.

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Delete 3.1.4 Sieve test entirely and substitute the following:

"3.1.4 Alternate tests. As an alternate to 3.1.1, 3.1.2 and 3.1.3, assays for zinc undecylenate and free undecylenic acid may be determined by the following tests:

"3.1.4.1 Assay for zinc undecylenate. The zinc undecylenate content shall be not less than 90.0 percent and not more than 110.0 percent of the labeled amount of zinc undecylenate under assayed as specified in 4.5.4.

"3.1.4.2 Assay for free undecylenic acid. The free undecylenic acid content shall be not less than 90.0 percent and not more than 125.0 percent of the labeled amount of free undecylenic acid when assayed as specified in 4.5.4."

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In 4.3.1 For end item testing, delete "Sieve test" and substitute "Alternate tests" with an inspection level of S-2 and a minimum sample of 5-composite.

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Add the following new paragraphs:

"4.5.4 Alternate tests.

"4.5.4.1 Assay for zinc undecylenate.

Standard preparation:

Pipet 1 and 3 ml of zinc stock solution (1000 ug/ml) into separate 100 ml volumetric flasks. Add 4 ml of 2.5 percent W/V hydrochloric acid and bring to the mark with purified water. The solutions contain 10 and 30 µg/ml zinc respectively.

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Sample preparation:

Accurately weigh approximately 1.5 gram of the powder into a 150 ml beaker. Add 60 ml of 2.5 percent w/v hydrochloric acid and heat on a hot plate until the solution begins to boil. Cool the solution and filter into a 200 ml volumetric flask and dilute to volume with purified water. Dilute 10 ml of this solution to 100 ml with purified water.

Procedure:

Alternately aspirate sample and standard solutions into a suitable atomic absorption spectophotometer according to the method outlined for the test atomic absorption spectophotometer for zinc analysis.

Calculation:

$$(\mu g/ml \text{ lower std} + \left[\frac{(\text{Sample reading}) - (\text{lower std reading})}{(\text{Upper std reading} - (\text{lower std reading})}\right]$$

$$[\mu g/ml \text{ upper std} - \mu g/ml \text{ lower std}] = \mu g/ml \text{ zinc in sample.}$$

$$\frac{(\mu g/ml \text{ zinc in sample}) \times 200 \times 10 \times 6.606 \times 100 \times 100}{(\text{sample weight in mg}) \times 1000 \times 20} = \text{percent label}$$

4.5.4.2 Assay for free undecylenic acid.

Internal standard preparation.

Dissolve 1.0 gram of tridecanoic acid in 100 ml of hexane.

Undecylenic acid standard preparation:

Accurately weigh 200 mg of undecylenic acid reference standard into a tared 100 ml beaker. Add 50 ml of hexane and dissolve the sample. Quantitatively transfer the solution into 200 volumetric flask, dilute to volume with hexane and mix well. Pipet 20.0 ml of this stock solution into a 50 ml volumetric flask containing 5.0 ml of the internal standard solution. Dilute to volume with hexane and mix well.



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Procedure:

Pipet 1.0 ml of the standard and sample solutions into separate 5 ml centrifuge tube. Evaporate to dryness without healing. Add 4 µl of Regisil and 1 ml of chloroform, stopper, mix and allow to stand for 10 minutes. Alternately inject 2 µl of the sample and standard solutions into a suitable gas chromatograph with the following chromatographic conditions:

Column - 3 percent SE - 30 on 100/200 mesh Supelcoport; 5.5 foot x 2 mm stainless steel.

Instrument - Varian Model 1200 or equivalent

Carrier gas - Helium, flow at 30 ml/min

Detector - Flame ionization

Flame gases - Hydrogen, 30 ml/min
Air, 300 ml/min

Column temperature - 165°C (isothermal)

Detector temperature - 250°C

Injection temperature - 250°C

Calculations:

$$\frac{(R\ \text{sample}) \times Cs \times 50 \times 100 \times 100}{(R\ \text{std}) \times \text{Wt of sample in mg} \times 2} = \text{percent label}$$

Where:

R sample - ratio of peak height (area) of undeylenic acid to tridecanoic acid in the sample

R std - ratio of peak height (area) of undeylenic acid to tridecanoic acid in the standard.

Cs - mg of standard per ml



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5.2.2.3 Add to end of paragraph:

"As an alternate six (6) immediate containers shall be packaged in a paperboard carrier type container (cluster pack) in such a manner so as to prevent spilling of contents unless the carrier is damaged or opened. The carrier shall have a lock type closure so designed to hold the contents securely in place."

5.3.1 Add the following at the end of first sentence:

"When the carrier containers (cluster packs) are used and are stacked in two or more tiers, a corrugated fiberboard pad shall be inserted between each tier.

5.3.2 Add the following at the end of first sentence:

"When the carrier containers (cluster packs) are used and are stacked in two or more tiers, a corrugated fiberboard pad shall be inserted between each tier.

Preparing activity:
Defense Logistics Agency - DM

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